

5.5.1 Stars

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the terms planets, planetary satellites, comets, solar systems, galaxies and the universe	HSW7
(b) formation of a star from interstellar dust and gas in terms of gravitational collapse, fusion of hydrogen into helium, radiation and gas pressure	
(c) evolution of a low-mass star like our Sun into a red giant and white dwarf; planetary nebula	HSW8
(d) characteristics of a white dwarf; electron degeneracy pressure; Chandrasekhar limit	HSW8
(e) evolution of a massive star into a red super giant and then either a neutron star or black hole; supernova	HSW8
(f) characteristics of a neutron star and a black hole	HSW8
(g) Hertzsprung–Russell (HR) diagram as luminosity-temperature plot; main sequence; red giants; super red giants; white dwarfs.	HSW8